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Development of Machine Learning–based Algorithms to Predict the 2- and 5-year Risk of TKA After Tibial Plateau Fracture Treatment

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Abstract

Background When faced with a severe intraarticular injury like a tibial plateau fracture, patients count on surgeons to make an accurate estimation of prognosis. Unfortunately, there are few tools available that enable precise, personalized prognosis estimation tailored to each patient's unique circumstances, including their individual and fracture-specific characteristics. In this study, we developed and validated a clinical prediction model using machine-learning algorithms for the 2- and 5-year risk of TKA after tibia plateau fractures.

Questions/purposes Can machine learning–based probability calculators estimate the probability of 2- and 5-year risk of conversion to TKA in patients with a tibial plateau fracture?

Methods A multicenter, cross-sectional study was performed in six hospitals in patients treated for a tibial plateau fracture between 2003 to 2019. In total, 2057 patients were eligible for inclusion and were sent informed consent and a questionnaire to inquire whether they underwent conversion to TKA. For 56% (1160 of 2057), status of conversion to TKA was accounted for at a minimum of 2 years, and 53% (1082 of 2057) were accounted for at a minimum of 5 years. The mean follow-up among responders was 6 ± 4 years after injury. An analysis of nonresponders found that responders were slightly older than nonresponders

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Ethical approval for this study was obtained from the University Medical Center Groningen (number: 201800411).
This work was performed at the University Medical Center Groningen, Groningen, The Netherlands.

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(53 ± 16 years versus 51 ± 17 years; p = 0.001), they were more often women (68% [788 of 1160] versus 58% [523 of 897]; p = 0.001), they were treated nonoperatively less often (30% [346 of 1160] versus 43% [387 of 897]; p = 0.001), and they had larger fracture gaps (6.4 ± 6.3 mm versus 4.2 ± 5.2 mm; p < 0.001) and step-offs (6.3 ± 5.7 mm versus 4.5 ± 4.7 mm; p < 0.001). AO Foundation/Orthopaedic Trauma Association (AO/OTA) fracture classification did not differ between nonresponders and responders (B1 11% versus 15%, B2 16% versus 19%, B3 45% versus 39%, C2 6% versus 8%, C3 22% versus 17%; p = 0.26). A total of 70% (814 of 1160) of patients were treated with open reduction and internal fixation, whereas 30% (346 of 1160) of patients were treated nonoperatively with a cast. Most fractures (80% [930 of 1160]) were AO/OTA type B fractures, and 20% (230 of 1160) were type C. Of these patients, 7% (79 of 1160) and 10% (109 of 1082) underwent conversion to a TKA at 2- and 5-year follow-up, respectively. Patient characteristics were retrieved from electronic patient records, and imaging data were shared with the initiating center from which fracture characteristics were determined. Obtained features derived from follow-up questionnaires, electronic patient records, and radiographic assessments were eligible for development of the prediction model. The first step consisted of data cleaning and included simple type formatting and standardization of numerical columns. Subsequent feature selection consisted of a review of the published evidence and expert opinion. This was followed by bivariate analysis of the identified features. The features for the models included: age, gender, BMI, AO/OTA fracture classification, fracture displacement (gap, step-off), medial proximal tibial alignment, and posterior proximal tibial alignment. The data set was used to train three models: logistic regression, random forest, and XGBoost. Logistic regression models linear relationships, random forest handles nonlinear complexities with decision trees, and XGBoost excels with sequential error correction and regularization. The models were tested using a sixfold validation approach by training the model on data from five (of six) respective medical centers and validating it against the remaining center that was left out for training. Performance was assessed by the area under the receiver operating characteristic curve (AUC), which measures a model's ability to distinguish between classes. AUC varies between 0 and 1, with values closer to 1 indicating better performance. To ensure robust and reliable results, we used bootstrapping as a resampling technique. In addition, calibration curves were plotted, and calibration was assessed with the calibration slope and intercept. The calibration plot compares the estimated probabilities with the observed probabilities for the primary outcome. Calibration slope evaluates alignment between predicted probabilities and observed outcomes (1 = perfect, < 1 = overfit, > 1 = underfit). Calibration intercept indicates bias (0 = perfect, negative = underestimation, positive = overestimation). Last, the Brier score, measuring the mean squared error of predicted probabilities (0 = perfect), was calculated.

Results There were no differences among the models in terms of sensitivity and specificity; the AUCs for each overlapped broadly and ranged from 0.76 to 0.83. Calibration was most optimal in logistic regression for both 2- and 5-year models, with slopes of 0.82 (random forest 0.60, XGBoost 0.26) and 0.95 (random forest 0.85, XGBoost 0.48) and intercepts of 0.01 for both (random forest 0.01 to 0.02; XGBoost 0.05 to 0.07). Brier score was similar between models varying between 0.06 to 0.09. Given that its performance metrics were highest, we chose the logistic regression algorithm as the final prediction model. The web application providing the prediction tool is freely available and can be accessed through: https://3dtrauma.shinyapps.io/tka_prediction/.

Conclusion In this study, a personalized risk assessment tool was developed to support clinical decision-making and patient counseling. Our findings demonstrate that machine-learning algorithms, particularly logistic regression, can provide accurate and reliable predictions of TKA conversion at 2 and 5 years after a tibial plateau fracture. In addition, it provides a useful prognostic tool for surgeons who perform fracture surgery that can be used quickly and easily with patients in the clinic or emergency department once it complies with medical device regulations. External validation is needed to assess performance in other institutions and countries; to account for patient and surgeon preferences, resources, and cultures; and to further strengthen its clinical applicability.

Level of Evidence Level III, therapeutic study.

Introduction

Lower leg injuries have a major impact on patient's mobility, social activities, and ability to work. In particular, injuries to the load-bearing part of the knee, including tibial plateau fractures, may substantially impair movement [10]. The incidence of tibial plateau fractures is 10 to 25 per 100,000 people [9, 12], which means that this injury affects about 2 million individuals worldwide each year. Especially in cities, where millions of people ride motorcycles, scooters, and bicycles, these injuries are common, and many are treated surgically. But despite achieving anatomic reduction, damage to the tibial plateau might progress to posttraumatic osteoarthritis and instability, resulting in pain and stiffness, and sometimes to a conversion to TKA [17, 18]. Compared with the general population, tibial plateau fractures are associated with a 3.5- to 5-times increased risk of TKA [8, 32].

Patients with these injuries often face difficult decisions about their treatment options, and shared decision-making is (or should be) a crucial aspect of their care. Patients may reasonably harbor uncertainty about injury severity, benefits and harms of surgery, and prognosis. However, estimating prognosis is challenging because so many factors can influence patients' results after these injuries, including patient characteristics (such as age, gender, BMI, comorbidities), fracture pattern, and the alignment of the fracture on preoperative and postoperative radiographs [1, 26, 32]. With so many variables to consider, accurate prediction of clinical outcomes after treatment remains challenging. Moreover, estimating prognosis often is subjective and relies heavily on the surgeon's experience as much or more than data-driven insights. Machine learning–based algorithms have the potential to combine different features into one model and show great potential in terms of fracture detection, grading osteoarthritis, and predicting outcomes like mortality [6, 13, 23]. These algorithms could be valuable as well in predicting clinical outcome after injuries. Such a model could benefit patient counseling and shared decision-making, as it would allow for more personalized and informed treatment plans. However, currently we lack good clinical prediction models in orthopaedic trauma surgery. Existing outcome prediction models in this field are limited to mortality or length of hospital stay [20, 25]. We therefore present an innovative proof of concept for developing clinical prediction models in orthopaedic trauma, aiming to establish a baseline work for outcome prediction after injuries. For this purpose, we collected clinical outcomes from a large, multicenter cohort of patients with tibial plateau fractures. Our goal was to develop and internally validate a machine learning–based prediction model utilizing patient characteristics for a patient-specific risk assessment of 2- and 5-year conversion to TKA. Additionally, we aimed to integrate the algorithm into an online, freely accessible prediction tool that can eventually be utilized at the point of care once it complies with medical device regulations.

In this study, we asked: Can machine learning–based probability calculators estimate the probability of 2- and 5-year risk of conversion to TKA in patients with a tibial plateau fracture?

Patients and Methods

This study is reported following the Transparent Reporting of Multivariable Prediction Models for Individual Prognosis or Diagnosis Guideline (TRIPOD statement [Appendix 1; http://links.lww.com/CORR/B400]) and the *Journal of Medical Internet Research* Guidelines for Developing and Reporting Machine Learning Predictive Models in Biomedical Research [5, 16]. A study protocol was not published.

Study Design and Setting

We performed a multicenter, cross-sectional study in patients treated for tibial plateau fractures in the orthopaedic and trauma surgery department of six hospitals within The Netherlands and Belgium between 2003 to 2019. Hospitals consisted of two academic centers and four nonacademic centers in medium-sized cities.

Patients

We identified 2241 patients who had an available diagnostic CT scan and who were still alive according to the national population registry. Adult patients (≥ 18 years old) who were treated operatively or nonoperatively for their tibial plateau fracture were potentially eligible. From the identified patients, we excluded 0.2% (4 of 2241) because they required amputation, and we excluded 4% (89 of 2241) because of coexisting conditions complicating outcome measurement, including Parkinson disease, cerebrovascular accident, and paralysis. In addition, we excluded 3% (60 of 2241) of patients because their address was unknown, and we excluded 1% (31 of 2241) because they spoke a foreign language. Subsequently, 2057 patients were eligible for inclusion and were sent informed consent and questionnaires by posted mail to inquire whether they underwent conversion to TKA. In case of no response, a reminder was sent. Fifty-six percent (1160 of 2057) of patients responded at 6 ± 4 years after injury (Fig. 1). For fifty-six percent (1160 of 2057), status of conversion to TKA was accounted for at a minimum of 2 years, and 53% (1082 of 2057) were accounted for at a minimum of 5 years.

Analysis of Nonresponders

For the nonresponse analysis, we used an independent-samples t-test for continuous variables and a chi-square test for noncontinuous variables. Overall, nonresponders did not differ very much from responders, which tends to lend credibility to the analysis. The nonresponse analysis showed that nonresponders were slightly younger than responders (mean ± SD 51 ± 17 years versus 53 ± 16 years; $p = 0.001$), there was a smaller proportion of women (58% [523 of 897] versus 68% [788 of 1160]; $p = 0.001$), and they received nonoperative treatment more often (43% [387 of 897] versus 30% [346 of 1160]; $p = 0.001$). Knee radiographs and CT images of nonresponders

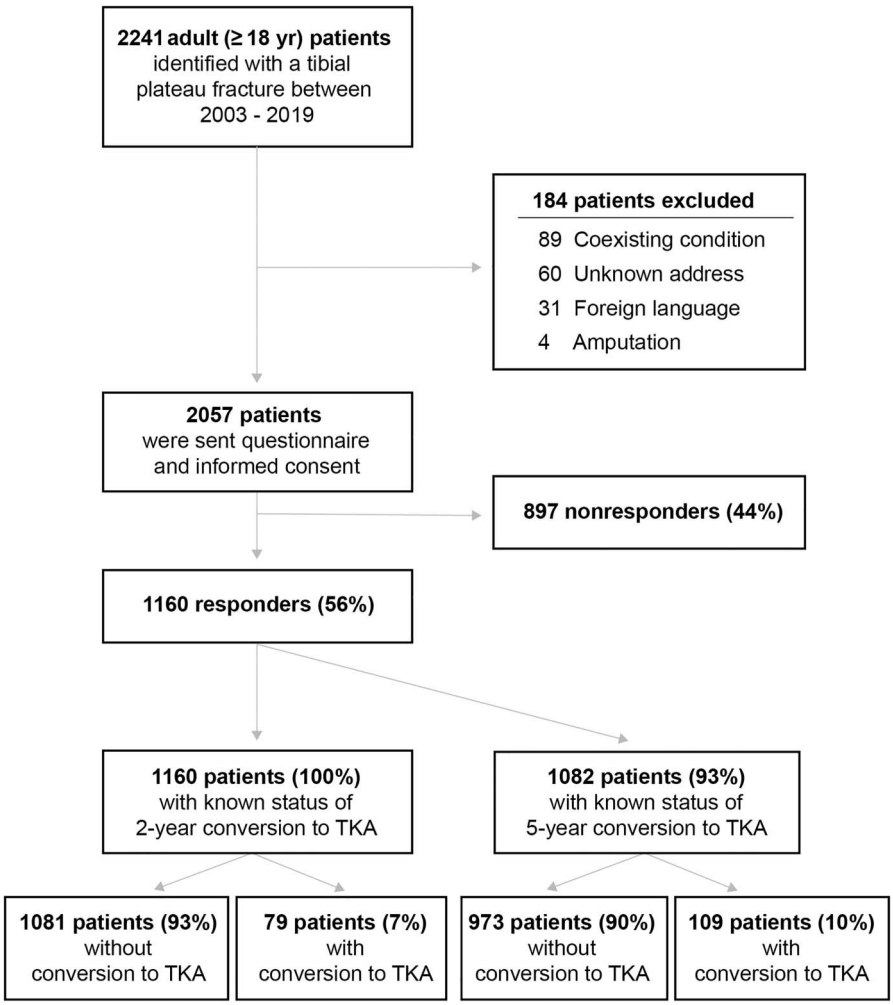


Fig. 1 The patient inclusion process for this study is depicted in this flow chart.

at the initiating center and one affiliated center were the only ones available for the imaging analysis (n = 550). Analysis showed that nonresponders in these centers had smaller gaps (4.2 ± 5.2 mm versus 6.4 ± 6.3 mm; $p < 0.001$) and step-off (4.5 ± 4.7 mm versus 6.3 ± 5.7 mm; $p < 0.001$). AO Foundation/Orthopaedic Trauma Association (AO/OTA) fracture classification did not differ between nonresponders and responders (B1 11% versus 15%, B2 16% versus 19%, B3 45% versus 39%, C2 6% versus 8%, C3 22% versus 17%; p = 0.26).

Variables and Data Sources

Patient characteristics were retrieved from electronic patient records and questionnaires, and imaging data were shared with the initiating center from which fracture characteristics were determined. Images from all patients were reassessed through consensus by two observers experienced in fracture assessment (NA, FFAIJ) to determine fracture severity, AO/OTA classification [22], and quality of achieved reduction. Fracture severity was assessed on preoperative CT scans by measuring gap and step-off (Supplemental Fig. 1; http://links.lww.com/CORR/B401). The quality of the operative reduction and tibial alignment was evaluated on postoperative radiographs by measuring articular reduction, medial proximal tibial angle (Supplemental Fig. 2A; http://links.lww.com/CORR/B401), posterior proximal tibial angle (Supplemental Fig. 2B; http://links.lww.com/CORR/B401), and condylar widening [1].

Descriptive Data

Enrolled patients were 53 ± 15 years old at the time of injury, most of whom (68% [788 of 1160]) were women (Table 1). A total of 70% (814 of 1160) of patients were

Table 1. Patient characteristics (n = 1160)

Characteristic	Value
Age in years	53 ± 15
Gender, men	32 (372)
BMI in kg/m^2	26 ± 4.7
Smoking	20 (234)
Comorbidities	
Diabetes	8 (92)
Cardiovascular disease	11 (125)
Rheumatoid arthritis	5 (59)
Fracture characteristics	
High-energy trauma	12 (136)
Side, right	39 (452)
AO/OTA classification	
41-B1	12 (138)
41-B2	24 (274)
41-B3	45 (518)
41-C1	4 (41)
41-C2	2 (19)
41-C3	15 (170)
Presence of coincident fibular head fracture	20 (228)
Initial gap in mm[a]	5 ± 6
Initial step-off in mm[a]	6 ± 6
Treatment	
Surgical	70 (814)
Nonsurgical	30 (346)
Radiographic outcome[a]	
Adequate condylar widening (0-5 mm)	77 (888)
Adequate medial proximal tibial angle (87° ± 5°)	84 (971)
Adequate posterior proximal tibial angle (9° ± 5°)	81 (938)
Adequate incongruence < 2 mm	61 (705)
Conversion to TKA (at any follow-up time point)	11 (128)

Data presented as mean ± SD or % (n).
[a]Supplemental Figs. 1 and 2 (http://links.lww.com/CORR/B401) provide an example of radiographic measurements.

treated with open reduction and internal fixation, whereas 30% (346 of 1160) of patients were treated nonoperatively with a cast. The majority of the fractures (80% [930 of 1160]) were AO/OTA type B fractures, whereas 20% (230 of 1160) of the fractures consisted of type C fractures. The average initial gap was 5 ± 6 mm, and the step-off 6 ± 6 mm. In total, 7% (79 of 1160) of patients had conversion to TKA at the 2-year follow-up, whereas this increased to 10% (109 of 1082) at the 5-year follow-up.

Primary Study Outcome

The primary outcome of the model was conversion to TKA at 2- and 5-year follow-up. Whether and when a patient underwent conversion to a TKA was collected by approaching all patients. These time points were selected because the conversion rate to TKA is particularly high within the first 5 years after fracture [11] and because of the availability of reliable data.

Data Cleaning and Feature Selection

Obtained features, derived from follow-up questionnaires, electronic patient records, and radiographic assessments, were eligible for development of the prediction model (Table 1). Data cleaning included simple type formatting (for example, converting "yes"/"no" columns into Booleans) and included standardization of numeric columns. Subsequent feature selection consisted of an evidence review according to the methodological framework principles as proposed by Levac et al. [14] and expert opinion. This was followed by bivariate analysis of the identified features. Following the rule of thumb for building prediction models with a binary outcome, we limited feature selection to ensure that at least 10 events (TKAs) for each feature were included within the model. Feature selection resulted in the following included features: age, gender, BMI, AO/OTA classification, initial gap, initial step-off, medial proximal tibial alignment (MPTA), and posterior proximal tibial alignment (PPTA).

Missing Data

There were no missing values other than for BMI. For BMI, 4% (52 of 1160) of patients had missing data. Missing values were imputed through K-nearest neighbors imputation. K-nearest neighbors imputation was chosen to increase biological feasibility of the imputations by basing them on similar cases.

Development and External Validation of the Model

We used the cleaned and processed data set to train three models: logistic regression, random forest, and XGBoost [3, 4]. To capture potential nonlinear relationships in the data, we incorporated restricted cubic splines into the logistic regression model, allowing for greater flexibility while avoiding overfitting. Logistic regression predicts the probability of an outcome based on a linear relationship between features and is valued for its simplicity and interpretability. In contrast, random forest combines multiple

decision trees to handle complex, nonlinear relationships, while XGBoost, a gradient boosting algorithm, builds models sequentially to correct errors and is renowned for its high performance and regularization capabilities. We chose these models because of their simplicity, which might boost performance given obtained data set size. Models were tested using sixfold cross-validation by training the model on data from five (of six) respective medical centers and validating it against the remaining center that was left out for training. To ensure optimal classification, every cross-validation fold used a subset of the training set to determine the classification threshold. This approach qualifies as external validation because the model's performance is assessed in sixfold on data from a distinct medical center that was not included during training, simulating its ability to generalize to unseen clinical environments.

Assessment of Model Performance

We used sixfold cross-validation to compare the models. Accuracy was deemed an inappropriate measure for performance given that the highly imbalanced nature of the data set allows an accuracy of 90% just from predicting the false label (that is, if the model always predicts no conversion to TKA, it will be right 90% of the time, as TKA within 5 years only occurred in 10% of the patients). Instead, we chose the area under the receiver operating characteristic curve (AUC) as the main performance metric. The AUC is widely used to measure accuracy of diagnostic tests and provides an aggregate measure of performance across all possible classification thresholds. The AUC score ranges from 0 to 1, with a score of 0.5 indicating random guessing and 1 indicating perfect performance. To ensure robust and reliable results, we used bootstrapping as a resampling technique to estimate the variability of the AUC. This method generates confidence intervals (CIs) by repeatedly sampling from the data set with replacement, helping to assess model stability and mitigate overfitting, especially when dealing with limited data. In addition, calibration curves were plotted, and calibration was assessed with the calibration slope and intercept. The calibration plot compares the estimated probabilities with the observed probabilities for the primary outcome. Calibration slope assesses how well predicted probabilities align with observed outcomes (1 = perfect, < 1 = overfitting, > 1 = underfitting), and the calibration intercept indicates overall prediction bias (0 = perfect, negative = underestimation, positive = overestimation). Last, the Brier score was calculated. The Brier score quantifies the accuracy of probabilistic predictions by measuring the mean squared difference between predicted probabilities and actual outcomes (0 = perfect).

Ethical Approval

We obtained ethical approval for this study from the University Medical Center Groningen, Groningen, The Netherlands. There was no patient and public involvement during design, conduct, reporting, interpretation, or dissemination of the study.

Statistical Analysis

Descriptive statistics were used to describe the study population. For normally distributed data, continuous variables have been presented as mean ± SD, and for nonnormally distributed data as median (IQR). Data preprocessing and analysis was performed using Python Scripting, version 3.10.6 (Python Software Foundation), importing the scikit-learn [27] and Pandas libraries [19].

Web Application

Algorithms that showed the highest performance metrics were integrated into an online, freely accessible prediction tool. Data are available upon request, whereas the code can be assessed online (https://github.com/RaulSantanaTrejo/TKA-Predictions-2025).

Results

Performance of Machine Learning–based Algorithms to Predict the Risk of Conversion to TKA in Patients With a Tibial Plateau Fracture at 2- and 5-Year Follow-up

There were no differences among the models for both 2- and 5-year follow-up in terms of sensitivity and specificity. The mean AUCs derived from the six folds of each model overlapped broadly and ranged from 0.76 to 0.83 (Fig. 2). AUCs ranging from the 0.7 to 0.9 are sufficient as a foundation for developing a robust model. After bootstrapping, the AUC in the random forest model was 0.79 (95% CI 0.73 to 0.83) and 0.80 (95% CI 0.76 to 0.85) for 2- and 5-year TKA, respectively. For the logistic regression model, these were 0.79 (95% CI 0.74 to 0.84) and 0.82 (95% CI 0.78 to 0.86), and for the XGBoost, these were 0.73 (95% CI 0.68 to 0.78) and 0.77 (95% CI 0.73 to 0.82) (Table 2).

Calibration of the 2-year TKA prediction models showed calibration slopes ranging from 0.26 to 0.82, intercepts ranging from 0.01 to 0.05, and Brier scores between 0.06 to 0.07. The 5-year models showed improved calibration with slopes from 0.48 to 0.95, intercept from 0.01 to 0.07, and Brier score of 0.08 to 0.09 (Table 2). The

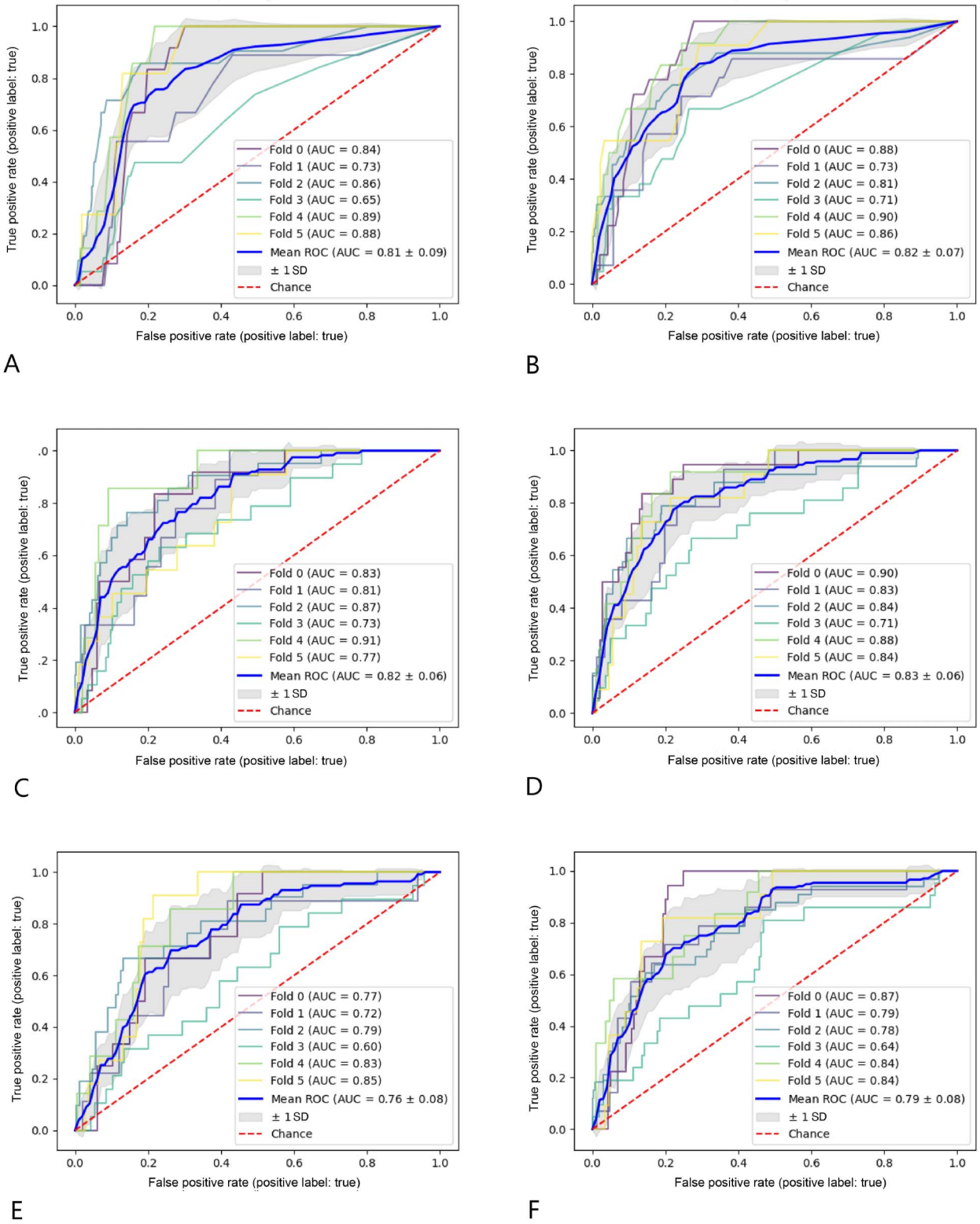


Fig. 2 ROC curves demonstrating the performances of the five folds of the (**A**) random forest 2-year, (**B**) random forest 5-year, (**C**) logistic regression 2-year, (**D**) logistic regression 5-year, (**E**) XGBoost 2-year, and (**F**) XGBoost 5-year algorithms for prediction of TKA. A color image accompanies the online version of this article.

Table 2. Performance of machine-learning algorithms predicting 2- and 5-year conversion to TKA following a tibial plateau fracture after sixfold validation by training the model on five (of six) hospitals and validating it on the excluded hospital

	Random forest		Logistic regression		XGBoost	
	2-year	5-year	2-year	5-year	2-year	5-year
AUC (95% CI)	0.79 (0.73 to 0.83)	0.80 (0.76 to 0.85)	0.79 (0.74 to 0.84)	0.82 (0.78 to 0.86)	0.73 (0.68 to 0.78)	0.77 (0.73 to 0.82)
Calibration slope (95% CI)	0.60 (0.46 to 0.74)	0.85 (0.73 to 0.97)	0.82 (0.66 to 0.98)	0.95 (0.82 to 1.07)	0.26 (0.16 to 0.36)	0.48 (0.38 to 0.58)
Calibration intercept (95% CI)	0.02 (0.01 to 0.04)	0.01 (-0.1 to 0.03)	0.01 (-0.01 to 0.03)	0.01 (-0.01 to 0.02)	0.05 (0.04 to 0.07)	0.07 (0.05 to 0.09)
Brier score	0.06	0.08	0.06	0.08	0.07	0.09

The AUC quantifies the ability of a model to distinguish between classes, with 1 indicating perfect discrimination and 0.5 representing random chance. The AUC was calculated following bootstrapping to correct for optimism. The calibration slope assesses how well predicted probabilities align with observed outcomes (1 = perfect, < 1 = overfitting, > 1 = underfitting), the calibration intercept indicates overall prediction bias (0 = perfect, negative = underestimation, positive = overestimation), and the Brier score quantifies the accuracy of probabilistic predictions by measuring the mean squared difference between predicted probabilities and actual outcomes (0 = perfect).

full calibration plots have been provided (Supplemental Fig. 3; http://links.lww.com/CORR/B402).

Developed Model

The logistic regression algorithm was chosen as the final prediction model because it outperformed the other models in terms of calibration, while the AUC was only marginally higher with substantial overlap across the models. Age, gap, and step-off were the strongest predictors in the final 2-year prediction model (Fig. 3), whereas age, BMI, and step-off were the strongest predictors in the final 5-year prediction model (Fig 4.).

Web Application

These algorithms were incorporated in a freely available, online prediction tool (three different vignettes that were imputed into the model have been provided [Fig. 5]). The web application providing the prediction tool can be accessed through https://3dtrauma.shinyapps.io/tka_prediction/.

Discussion

Despite achieving anatomic reduction, damage to the articular surface of the tibial plateau can progress to posttraumatic osteoarthritis and instability, leading to pain, stiffness, and, in some instances, patients' query for conversion to TKA. Patients with these injuries often face complex decisions about their index treatment options, making shared decision-making a crucial aspect of their care. However, with numerous variables to consider, accurately predicting clinical outcomes after treatment remains a significant challenge. Prognosis estimation is often flawed by surgeon biases rather than data-driven insights. Currently, such robust clinical prediction models in orthopaedic trauma surgery to assist in shared decision-making are scarce. In this study, we developed a machine learning–driven model that provides personalized outcome predictions for each respective patient based on specific patient and fracture characteristics. This approach advances prior research that primarily focused on associations between individual features and the risk of TKA [1, 2, 30]. Our model predicts the likelihood of conversion to TKA following a tibial plateau fracture and is integrated into an easy-to-use web application. The tool is designed with robust modeling and is intended at future use at the point of care, but it must first comply with medical device regulations. Further validation will be critical for further strengthening its clinical applicability.

Limitations

Our developed model was compromised by several limitations that arose from the study setup. First, selection bias caused by loss to follow-up of nonresponders is inherent to the cross-sectional study design. By contacting all patients multiple times, we minimized the risk of nonresponse bias as much as possible. Our overall response rate of 56% aligns with expectations for postal surveys based on existing evidence [21]. Nonresponse analysis showed that nonresponders were predominantly patients with relatively simpler and less displaced fractures with potentially lesser

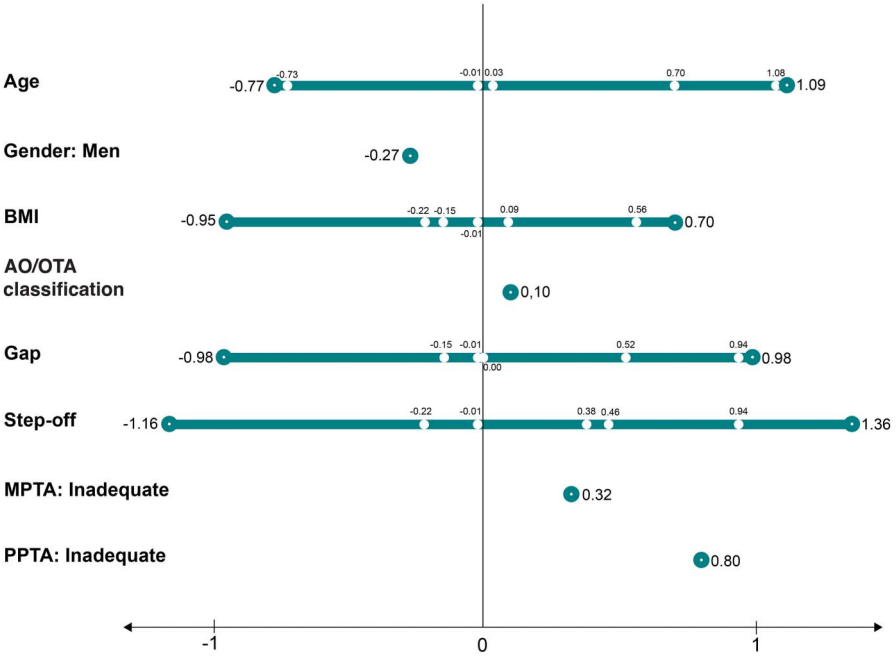


Fig. 3 Weights of the different included features within the final logistic regression model to predict 2-year conversion to TKA.

complaints, which could explain their lower participation rate. In the context of our study, this means that there is a risk that our model will produce positively biased estimates of probabilities (that is, to overestimate the risk of TKA). Furthermore, the retrospective character of the data set was a limitation because we had to handle missing data. However, missing values among the included model features were minimal, limited to BMI in only 4% of patients'

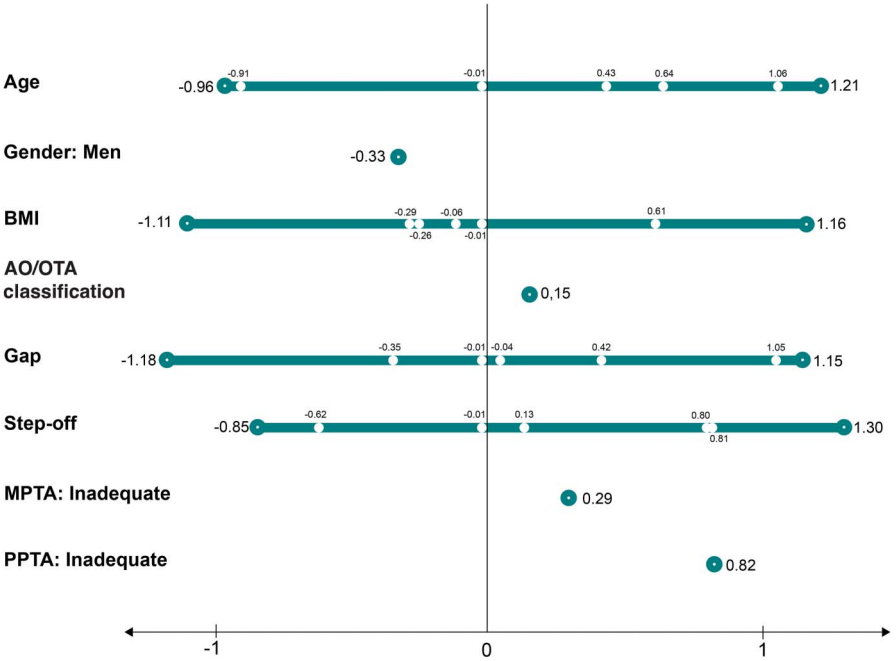


Fig. 4 Weights of the different included features within the final logistic regression model to predict 5-year conversion to TKA.

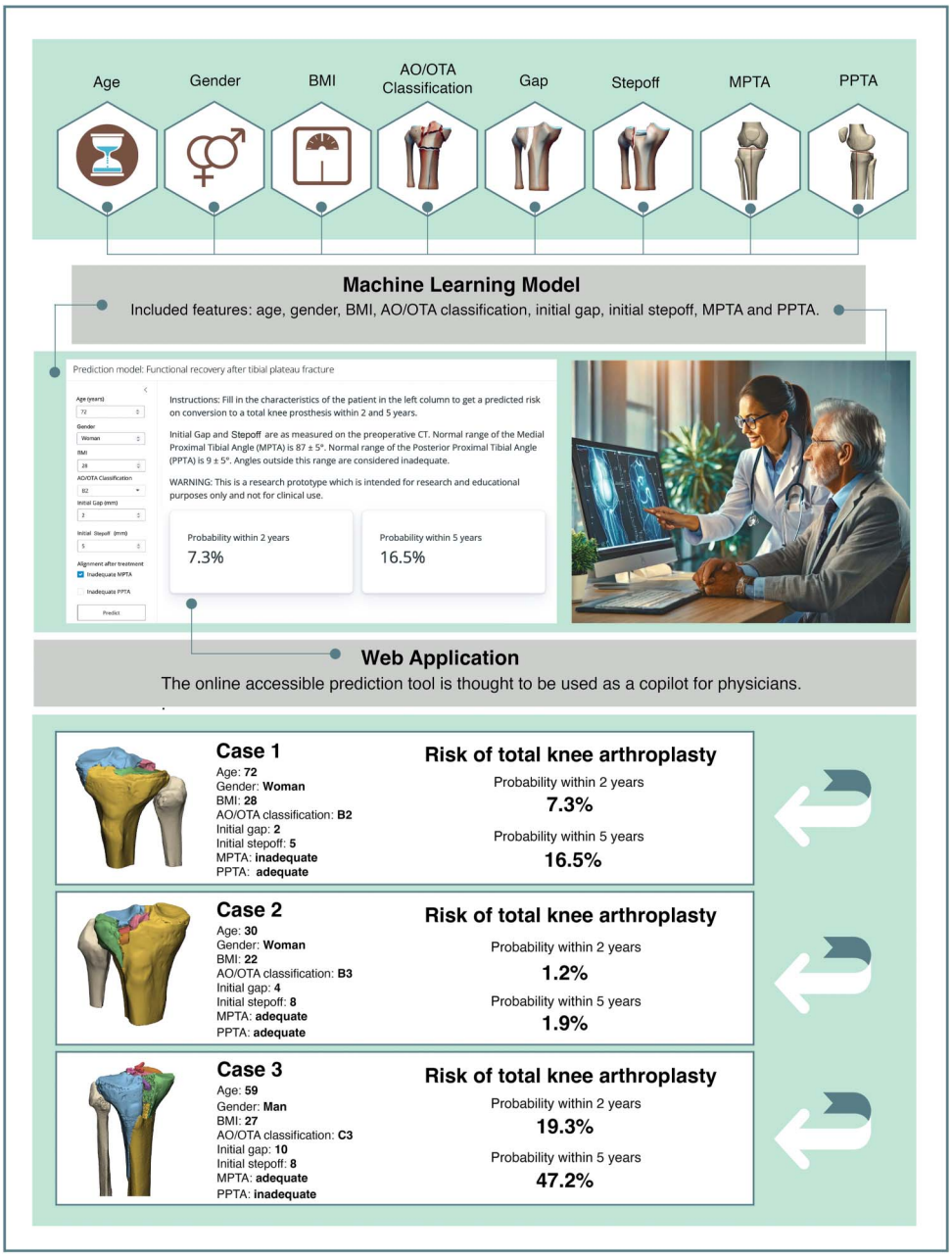


Fig. 5 Graphical illustration of the potential clinical usage of the machine learning–based prediction model. The included features (age, BMI, AO/OTA classification, gap, step-off, MPTA, and PPTA) can be imputed for each patient into the model to compute a personalized risk of TKA at 2- and 5-year follow-up. A total of three clinical vignettes are presented to illustrate different risks for these individual patients. The first involved a 72-year-old woman with a BMI of 28 kg/m^2. After falling from the stairs, she suffered an AO/OTA B2 fracture with a 2-mm gap and a 5-mm step-off. After treatment, MPTA remained inadequate, whereas PPTA was restored. The model estimated the risk of conversion to a TKA at 7.3% in 2 years and 16.5% in 5 years. The second was a 30-year-old woman with a BMI of 22 kg/m^2. The patient was involved in a motor vehicle accident, resulting in an AO/OTA B3 tibial plateau fracture with a gap of 4 mm and a step-off of 8 mm. Surgical treatment resulted in adequate MPTA and PPTA, meaning that the risk of this patient receiving a TKA is only 1.2% and 1.9% at the 2- and 5-year follow-up, respectively. The last vignette involved a 59-year-old man who fell from his bicycle. The man, with a BMI of 27 kg/m^2, sustained a C3 tibial plateau fracture with a gap and step-off of 10 and 8 mm, respectively. After surgical treatment, MPTA was restored, but the PPTA remained inadequate. This would give the man a risk of 19.3% of conversion to a TKA in 2 years, and 47.2% in 5 years according to the model. A color image accompanies the online version of this article.

data. Despite these gaps, our best-performing model demonstrated high AUC and strong calibration, suggesting that the missing potential features had a limited impact on performance. Nonetheless, future model improvements could benefit from incorporating baseline osteoarthritis status and mental health data. In addition, the follow-up period was limited to the midterm, with conversion to TKA assessed at endpoints of 2 and 5 years. As evidence indicates that most conversions to TKA happen within this timeframe [11], our model focusses on the midterm outcome. However, given that osteoarthritis tends to progress over time, it may be valuable for future models to incorporate the risk of long-term outcomes (> 10 years). Last, our model was validated using sixfold external validation, where in each fold, one hospital was excluded from the training set and used as an external validation set. While this approach ensured an external validation process, the six hospitals were all located within a specific geographical region (northwest Europe) with similar healthcare systems. This regional limitation may restrict the generalizability of the model. Future external validation should include data from diverse regions and international healthcare systems to more comprehensively evaluate the model's robustness and applicability.

Discussion of Key Findings

In this study, we developed a personalized risk assessment tool using patient and fracture characteristics to support clinical decision-making and patient counseling. Our findings demonstrate that machine learning–driven probability calculators, particularly logistic regression, can provide accurate and reliable predictions of TKA conversion at 2 and 5 years after a tibial plateau fracture. In a clinical setting, this model (freely available at https://3dtrauma.shinyapps.io/tka_prediction/) is intended for use in shared decision-making, which eventually could result in a better informed patient whose expectations regarding expected outcome could be better managed. However, it must first meet medical device regulations before being approved for clinical use. Additionally, this prediction model might also guide discussions between professionals regarding surgical treatment plans, including considering primary TKA or minimizing surgical exposure as a bridge to secondary TKA in patients with a high probability of conversion to TKA after osteosynthesis [28, 29]. While our model focuses on predicting clinical outcomes, most existing outcome prediction models in orthopaedic trauma are limited to mortality or length of hospital stay [7]. One recent study developed a machine-learning model to predict 2-year conversion to arthroplasty after a femoral neck fracture but performed poorly with AUCs of 0.57 to 0.64 and calibration slopes of 0.53 to 0.82 [31]. Another study sought to predict the risk of total knee replacement within 5 years in patients presenting with recent knee pain and showed good discrimination and calibration with an AUC of 0.79 to 0.87 and calibration slope of 0.95 to 0.96 [15]. Similarly, our best-performing models achieved AUCs of 0.79 and 0.82 for the 2- and 5-year predictions, respectively, with calibration slopes of 0.82 and 0.95, indicating comparable performance to the previously mentioned model. This model forms a reasonable basis for making prognostic estimates; however, external validation across diverse regions and healthcare systems is necessary to fully assess its robustness and applicability before implementation.

Our study showed that a relatively simple model such as logistic regression performed better than more complex models such as XGBoost. Simpler models are usually more resistant to issues like relatively small sample size, whereas complex models are prone to overfitting on small data sets [24]. The addition of the cubic splines further improved the model by more flexible relationships between the predictor variables and the outcome, particularly capturing nonlinear effects of age or BMI where the impact on the outcome may vary at different stages. Our final logistic regression model delivers robust and reliable predictions despite the relatively small data set. Also, by using these explainable models, physicians can eventually make better treatment decisions based on data. When comparing our data set with different prediction models in orthopaedic trauma research, our model is based on more patients than the median sample size of 875 within these studies [7], indicating that, despite the relatively small data set, it is among the larger data sets in orthopaedic trauma research. Future refinement of the model could involve the creation of larger, prospectively collected data sets, which would enable the use of more complex models and improve the accuracy and predictive power of the tool. This would enhance its ability to capture nuanced relationships within the data and offer even more precise predictions for TKA conversion, although the personalized risk should always be interpreted by the physician and considered alongside professional experience and the patient's wishes.

Conclusion

In this study, we developed and validated a machine learning–based prediction model (https://3dtrauma.shinyapps.io/tka_prediction/) for identifying patients at risk of conversion to a TKA within 2 and 5 years after a tibial plateau fracture. Using patient and fracture characteristics, we developed a personalized risk assessment tool to support clinical decision-making and patient counseling, and we believe that integrating machine learning into clinical practice can help manage expectations and enable

personalized treatment plans. Our results indicate that machine-learning algorithms, particularly logistic regression, can achieve reasonable accuracy and performance in predicting TKA conversion. Future studies should focus on further refinement, external validation, and compliance with medical device regulations of these models to improve their accuracy and reliability before they can be definitively implemented in clinical practice.

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